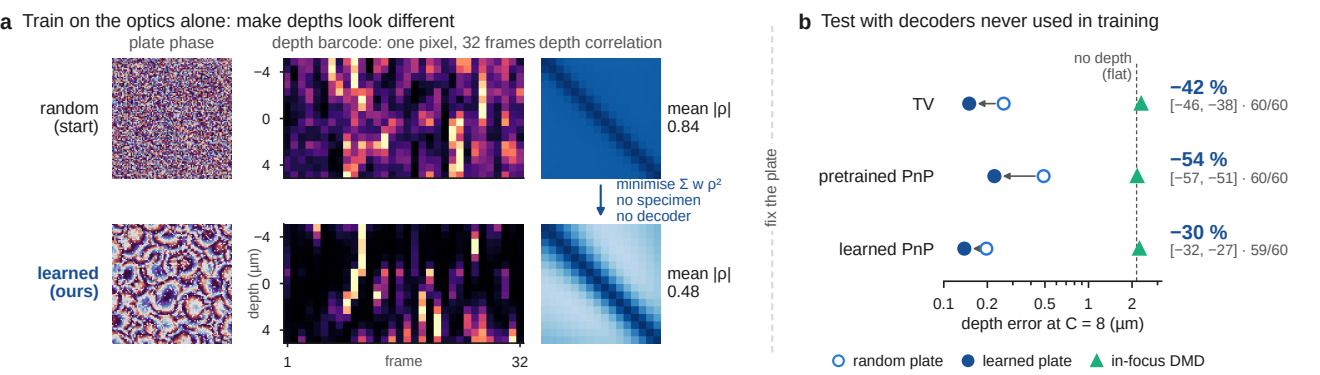

a
Train on the optics alone: make depths look different
plate phase
depth barcode: one pixel, 32 frames
depth correlation
random (start)
learned (ours)
−4
0
4
depth (μm)
1
frame
32
mean |ρ| 0.84
mean |ρ| 0.48
minimise Σ w ρ²
no specimen
no decoder
fix the plate
b
Test with decoders never used in training
no depth (flat)
TV
pretrained PnP
learned PnP
−42 %
[−46, −38] · 60/60
−54 %
[−57, −51] · 60/60
−30 %
[−32, −27] · 59/60
0.1
0.2
0.5
1
2
depth error at C = 8 (μm)
random plate
learned plate
in-focus DMD